Combined Science: Trilogy, Biology Paper 1 topics, Foundation Tier

Mark scheme

Using this mark scheme. A slash (/) separates alternative answers that each score. Words in brackets are not needed for the mark. Give the mark for any answer that means the same as the one listed. In calculations, give full marks for the correct answer with no working, and the working mark if the answer is wrong but the method is right. The extended response (06.1) is marked in levels.

Question 01

Question	Answers	Extra information	Mark
01.1	Chloroplast	Only one box ticked. Two or more ticked: 0 marks.	**1**
01.2	6 ÷ 0.06 (1) = (×) 100 (1)	Correct answer with no working: 2 marks. Do not accept 0.01.	**2**
01.3	Any two from: the cell grows / gets bigger (1) the DNA is copied / the chromosomes are copied (1) the number of sub-cellular structures increases, such as ribosomes or mitochondria (1)	Do not accept "the nucleus divides": that is mitosis itself.	**2**
	Total Question 01		**5**

Question 02

Question	Answers	Extra information	Mark
02.1	(2.80 - 2.50) = 0.30 and 0.30 ÷ 2.50 × 100 (1) = (+) 12 (%) (1)	Correct answer with no working: 2 marks. Accept +12.0. Do not accept -12.	**2**
02.2	0.2 and 0.4 (mol/dm^3)	Both needed, in either order.	**1**
02.3	Water moved out of the potato (cells) by osmosis (1) because the solution was more concentrated than the inside of the potato cells / the potato cells were more dilute (1) water moves from a dilute solution to a more concentrated solution through a partially permeable membrane / through the cell membrane (1)	Do not accept "sugar moved in" or "the potato lost sugar".	**3**
02.4	The cylinders had different starting masses / so the results can be compared (fairly)	Do not accept "it's more accurate" on its own.	**1**
	Total Question 02		**7**

Question 03

Question	Answers	Extra information	Mark
03.1	Lipase (1) fatty acids and glycerol (1)	Both products needed for the second mark. Do not accept "fats" or "amino acids".	**2**
03.2	The active site changes shape / the enzyme is denatured (1) so the substrate no longer fits (into the active site) (1)	Do not accept "the enzyme is killed" or "the enzyme dies": no marks for that idea.	**2**
03.3	Biuret (reagent / solution) (1) purple / lilac / mauve (1)	Ignore the colour before the test (blue). Do not accept "pink" on its own.	**2**
	Total Question 03		**6**

Question 04

Question	Answers	Extra information	Mark
04.1	It keeps the coronary artery open / widens the (narrowed) coronary artery (1) so blood (carrying oxygen) can flow to the heart muscle (1)	"Artery" without "coronary" can get the first mark if the answer is clearly about the heart. Do not accept "unblocks a vein".	**2**
	Total Question 04		**2**

Question 05

Question	Answers	Extra information	Mark
05.1	Antibiotics only kill bacteria / antibiotics don't kill viruses / viruses live inside cells (where antibiotics can't reach them)	Do not accept "the virus is resistant".	**1**
05.2	A small amount of dead / inactive (form of the) measles virus is injected (1) white blood cells make antibodies (against it) (1) if the real virus gets in later, white blood cells make the right antibodies quickly (and in large numbers), so the virus is destroyed before the person becomes ill (1)	Accept "the body remembers" for the third mark only if it's linked to antibodies being made faster. Do not accept "the vaccine contains antibodies".	**3**
	Total Question 05		**4**

Question 06

Question	Answers	Extra information	Mark
06.1	Level of response marking: see below.		**6**
	Total Question 06		**6**

Question 06.1: levels of response

Level	Descriptor
Level 3: 5 to 6 marks	A clear method in a logical order that would work. It says how light intensity is changed, how the rate of photosynthesis is measured, and at least two things kept the same, with a repeat or mean.
Level 2: 3 to 4 marks	A method that would mostly work, but with a step missing or unclear. It changes light intensity and measures something, and mentions at least one thing kept the same.
Level 1: 1 to 2 marks	Some relevant steps, but no clear order, or it doesn't say how the rate is measured.
0 marks	No relevant content.

Indicative content

- put the pondweed in a boiling tube or beaker of water (sodium hydrogencarbonate can be added to give plenty of carbon dioxide)
- put a lamp at a set distance from the pondweed, such as 10 cm, measured with a ruler
- leave it for a few minutes to settle before starting
- count the bubbles of oxygen given off in a set time, such as one minute, or collect the gas and measure its volume
- move the lamp to different distances, such as 20, 30, 40 and 50 cm, and repeat the count at each one
- repeat each distance and work out a mean
- keep the same: the temperature (a beaker of water between lamp and plant, or an LED bulb), the same pondweed, the amount of carbon dioxide, the time for each count, no other light

Total: 30 marks. As a rough guide for this practice paper only (not exam board grade boundaries): 22 or more is a strong result, 15 to 21 is solid with gaps to fix, under 15 means going back over these topics in the notebook first.